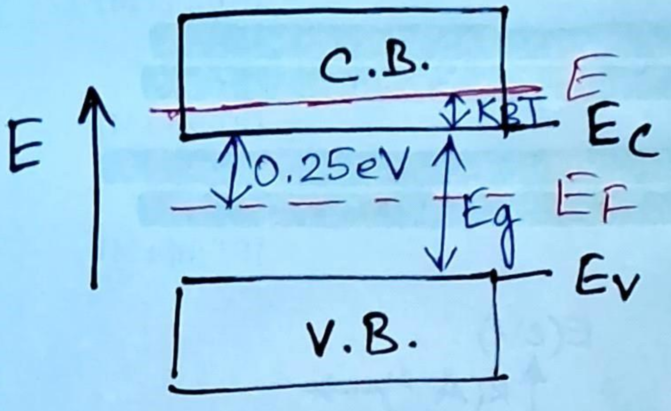
Problem on Fermi Dirac distribution in semiconductor



@ room temperature (300K)

$K_BT = \cancel{8}\ 26$ meV

(i) $E - E_C = K_BT$

$= 26$ meV — Ⓐ

(ii) $E_C - E_F = 0.25$ eV

$= 250$ meV — Ⓑ

Ⓐ + Ⓑ :

$$E - E_C + E_C - E_F = (26 + 250) \text{ meV}$$

$$E - E_F = 276 \text{ meV}$$

$$P_{occupied}(E) = P_{FD}(E)$$

$$= \frac{1}{1 + \exp\left(\frac{E - E_F}{K_BT}\right)}$$

$$= \frac{1}{1 + \exp\left(\frac{276 \text{ meV}}{26 \text{ meV}}\right)}$$

A quantum state in the conduction band is K_BT above conduction band minimum E_C of an intrinsic semiconductor. The Fermi level is 0.25eV below E_C. Find the probability that the quantum state is occupied at 300K.

Problem on equilibrium carrier concentration in i-s.c. @ room temperature (300K)

$$N_c = 2.8 \cdot 10^{19}\ cm^{-3}$$

$$n_0 = ?$$

$$n_0 = N_c \exp\left(-\frac{(E_c - E_F)}{k_B T}\right)$$

We have

$$E_c - E_F = 250\,meV$$

so

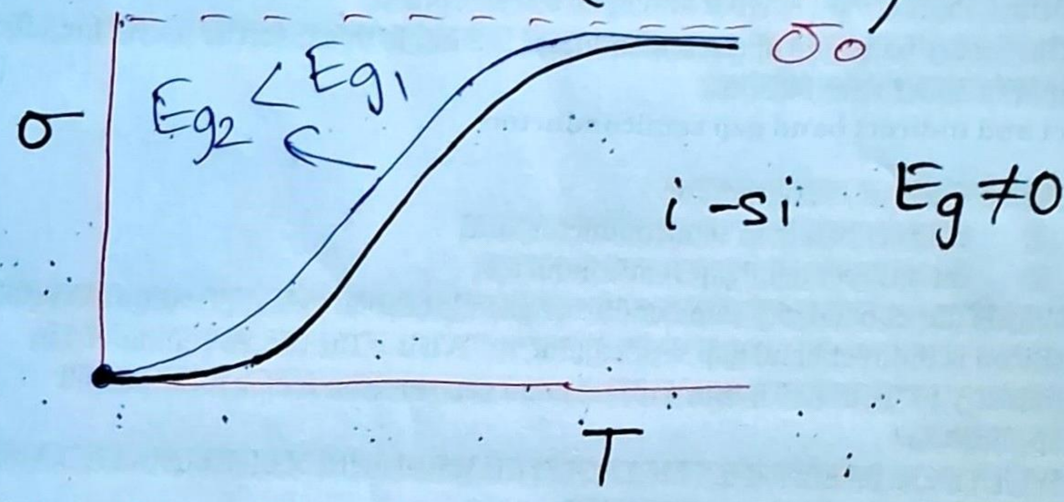
$$n_0 = 2.8 \cdot 10^{19}\ cm^{-3} \cdot \exp\left(-\frac{250\,meV}{26\,meV}\right)$$

$$= 2.8 \cdot 10^{19} \cdot \exp\left(-\frac{250}{26}\right) cm^{-3}$$

For the same material in the above problem, if the effective density of states in conduction band N_c is $2.8 \cdot 10^{19}\ cm^{-3}$ at 300K, then find the electron concentration.

Problem on conductivity vs energy band gap (Mock paper set 2, 1(b))

$$\sigma = \sigma_0 \exp\left(-\frac{E_g}{2k_BT}\right)$$



But if: $E_g = 0$

$$\sigma = \sigma_0 \exp\left(-\frac{0}{k_BT}\right)$$

$$= \sigma_0$$

~~But~~ ∴ for smaller E_g, we have more conductivity @ temperatures.

$$E_g(Si) = 1.12\,eV$$

$$E_g(Ge) = 0.66\,eV$$

$$\boxed{\begin{aligned} &\because E_g(Ge) < E_g(Si) \\ &\Rightarrow \sigma(Ge) > \sigma(Si) \end{aligned}}$$

Why Germanium has more conductivity than Si at all temperatures?

Problem on position of E_F (Mock paper set 2, 2(b))

$$E_g = 0.7\,eV$$

$$T = 300K \Rightarrow k_B T = 26\,meV$$

$$m_h^* = 2m_e^* \Rightarrow m_p^* = 2m_n^*$$

(h) hole ≡ positive (p)

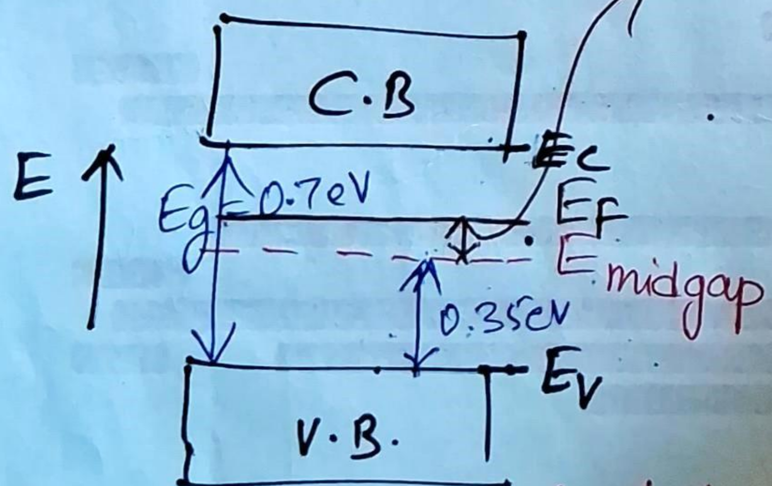
(e) electron ≡ negative (n)

$$E_F = E_{midgap} + \frac{3}{4} k_B T \ln \frac{m_p^*}{m_n^*}$$

$$= E_{midgap} + \frac{3}{4} 26\,meV \ln \frac{2m_n^*}{m_n^*}$$

$$= E_{midgap} + \frac{3}{4} 26\,meV \ln 2$$

$$E_F - E_{midgap} = \frac{3}{4} \cdot 26\,meV \ln 2 > 0$$



For an intrinsic semiconductor with a bandgap of 0.7eV, determine the position of E_F at 300K, if $m_h^* = 2m_e^*$.

Problem on position of E_F

$$T = 300K \Rightarrow k_B T = 26 \text{ meV}$$

density of states effective masses
$$\begin{cases} m_n^* = 1.08\, m_0 \\ m_p^* = 0.56\, m_0 \end{cases}$$

// Generally, conductivity effective masses of electron is greater than hole.

There is difference between DOS effective mass and conductivity effective mass (Advanced topic)

$$E_F = E_{midgap} + \frac{3}{4} k_B T \ln \frac{m_p^*}{m_n^*}$$

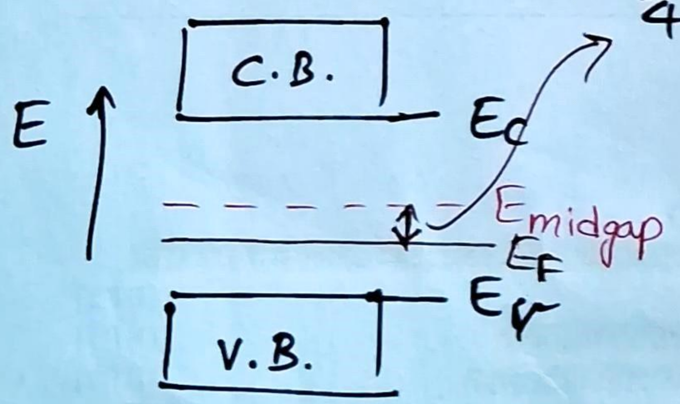
$$= E_{midgap} + \frac{3}{4} k_B T \ln \frac{0.56\, m_0}{1.08\, m_0}$$

$$= E_{midgap} + \frac{3}{4} k_B T \ln \frac{0.56}{1.08}$$

$$= E_{midgap} - \frac{3}{4} k_B T \ln \frac{1.08}{0.56} \qquad // \ln \frac{1}{x} = -\ln x$$

$$\therefore E_F - E_{midgap} = -\frac{3}{4} k_B T \ln \frac{1.08}{0.56}$$

$$= -\frac{3}{4} \cdot 26 \ln \frac{1.08}{0.56} \text{ meV}$$



Calculate the position of Fermi level with respect to the center of bandgap in Si at 300K. Take $m_n^* = 1.08\, m_0$ and $m_p^* = 0.56\, m_0$, where m_0 is free electron mass.

Problem on resistivity ~~of Ge at 300K~~ of i-s.c.

$n_i = 2.4 \cdot 10^{13}\ cm^{-3} = 2.4 \cdot 10^{19}\ m^{-3}$ //

// $1 m^3 = 10^6\ cm^3$

// $\Rightarrow \frac{1}{cm^3} = \frac{1}{10^{-6} m^3} = 10^{+6} m^{-3}$

$\mu_n = 3900\ cm^2 V^{-1} s^{-1} = 3.9 \cdot 10^3\ cm^2 V^{-1} s^{-1}$

$= 3.9 \cdot 10^{-1}\ m^2 V^{-1} s^{-1}$

// $1 m^2 = 10^4\ cm^2$

// $\Rightarrow 1 cm^2 = 10^{-4}\ m^2$

$\mu_p = 1900\ cm^2 V^{-1} s^{-1} = 1.9 \cdot 10^3\ cm^2 V^{-1} s^{-1}$

$= 1.9 \cdot 10^{-1}\ m^2 V^{-1} s^{-1}$

$\sigma_i = n_i e (\mu_n + \mu_p)$

$= 2.4 \cdot 10^{19} \cdot 1.6 \cdot 10^{-19} (3.9 \cdot 10^{-1} + 1.9 \cdot 10^{-1})$

$= 2.4 \cdot 1.6 \cdot 5.8 \cdot 10^{-1}\ S m^{-1}$

$= 2.4 \cdot 1.6 \cdot 5.8 \cdot 10^{-1}\ \Omega^{-1} m^{-1}$

$\therefore \rho_i = \frac{1}{\sigma_i} = \frac{10^1}{2.4 \cdot 1.6 \cdot 5.8}\ \Omega m$

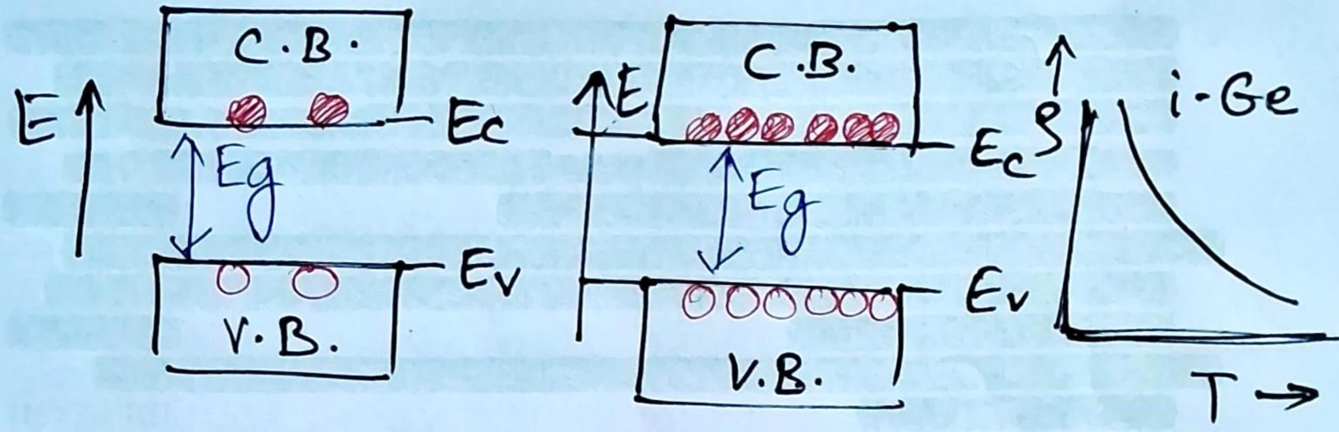
Find the resistivity of i-Ge @ 300K.

Take ~~$E_g(Ge) = 0.66 eV$~~ $n_i(Ge) = 2.4 \cdot 10^{13}\ cm^{-3}$

$\mu_n(Ge) = 3900\ cm^2 V^{-1} s^{-1}$

$\mu_p(Ge) = 1900\ cm^2 V^{-1} s^{-1}$

Problem on resistivity vs temperature of i-S.C.



low T $\Longrightarrow$ high T

(i) E_g is independent of temperature

(ii) With increase in temperature, more electron-hole pairs are generated, so that conductivity increases

(iii) Resistivity decreases, so that i-S.C. has negative temperature coefficient (NTC) for thermistor application

$$\sigma_i = \sigma_0 \exp\left(-\frac{E_g}{2k_BT}\right)$$

$$\Rightarrow \rho_i = \frac{1}{\sigma_i} = \frac{1}{\sigma_0 \exp\left(-\frac{E_g}{2k_BT}\right)} = \frac{1}{\sigma_0}\exp\left(\frac{E_g}{2k_BT}\right)$$

$// \frac{1}{e^x} = e^{-x}$

$$\rho_i = \rho_0 \exp\left(\frac{E_g}{2k_BT}\right) \quad // \rho_0 = \frac{1}{\sigma_0}$$

$$\therefore \frac{\rho_i(\overset{100^\circ C}{\cancel{300K}})}{\rho_i(300K)} = \frac{\rho_i(400K)}{\rho_i(300K)} = \frac{\exp\left(\frac{E_g}{2k_BT_{400K}}\right)}{\exp\left(\frac{E_g}{2k_BT_{300K}}\right)}$$

[contd...]

The resistivity of i-Si is 2.3·10 Ωm at 300K. Calculate its resistivity at 100°C.

[Take $E_g(Si) = 1.12eV$]

...contd] Now $K_B T_{400K} = K_B T_{300} \cdot \frac{T_{400K}}{T_{300}}$

$$= 26\,meV \cdot \frac{400}{300}$$

$$= 26 \cdot \frac{4}{3}\ meV$$

$$\therefore \frac{\rho_i(100°C)}{\rho_i(300K)} = \exp\left[\frac{E_g}{2}\left(\frac{1}{K_B T_{400K}} - \frac{1}{K_B T_{300K}}\right)\right]$$

// $\frac{\exp(x)}{\exp(y)} = \exp(x-y)$

$$\therefore \rho_i(100°C) = \rho_i(300K) \cdot \exp\left[\frac{E_g}{2}\left(\frac{1}{26 \cdot \frac{4}{3}\,meV} - \frac{1}{26\,meV}\right)\right]$$

Now $E_g = 1.12\ eV = 1120\ meV$

$\rho_i(300K) = 2.3 \cdot 10^3\,\Omega m$

$$\therefore \rho_i(100°C) = 2.3 \cdot 10^3\,\Omega m \cdot \exp\left(\frac{3}{4} \cdot \frac{560}{26} - \frac{560}{26}\right)$$

$$\rho_i(100°C) = 2.3 \cdot 10^3 \cdot \exp\left(-\frac{1}{4} \cdot \frac{560}{26}\right)\Omega m$$